

Requirement Analysis

Tools & Technologies

Date	
Team ID	6aba07eeb2817f097462be97
Project Name	FitBuddy – AI Fitness Plan Generator using Gemini Models
Minimum Marks	

Technology Stack & Tools Used:

FitBuddy uses a combination of modern web technologies and AI tools to provide personalized fitness planning. Gemini API acts as the core AI component for generating fitness plans and recommendations. React.js is used to create the interactive user interface, while Node.js handles the backend logic and API communication. MongoDB stores user profiles, workout plans, and progress data. Tailwind CSS provides a clean and responsive design, and Vercel is used to deploy the application for easy access.

Together, these technologies help FitBuddy deliver a personalized, responsive, and user-friendly AI fitness experience.

Tools, Technologies and Their Applications:

FitBuddy Technology Stack

The tools, languages and frameworks that power FitBuddy

Programming Languages

Language	Used for
 Python 3.10+	The whole backend: routes, database, AI calls, tests
 HTML5	Page structure (Jinja2 templates)
 CSS3	Styling and layout
 JavaScript	A small script that shows "Working with AI..." on the button
 SQL	Written for you by SQLAlchemy, so you don't write it yourself

AI Models

Model	Used for
 Google Gemini pro-tier model (default gemini-3.8-flash)	Generating the 7-day workout plan and rewriting it from user feedback
 Google Gemini Flash-Lite (default gemini-3.5-flash-lite)	Generating the short nutrition tip
 Fallback models (gemini-3.7-flash, gemini-3.6-flash, gemini-2.5-flash and others)	Tried automatically if the main model is overloaded

 These are pre-trained large language models used through prompts. We don't train any model ourselves.

Frameworks and Libraries

Tool	Purpose
 FastAPI	Web framework for the backend and API
 Uvicorn	Server that runs the app
 Jinja2	Fills the HTML templates with data
 Pydantic	Checks user input (age, weight, goal and so on)
 SQLAlchemy	Talks to the database
 SQLite	The database, stored in the fitbuddy.db file
 google-genai	Google's Python library for calling Gemini
 python-dotenv	Reads your API key from .env
 pytest and httpx	Automated testing

Development Tools

Tool	Purpose
 VS Code	Code editor
 Python venv	Keeps the project's libraries separate
 pip	Installs the libraries
 Google AI Studio	Where you get the Gemini API key
 Git	Optional version control (.gitignore is included)

 Better Tools → Smarter Development → FitBuddy 